Foundational Model for Rover Autonomy

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Introduction

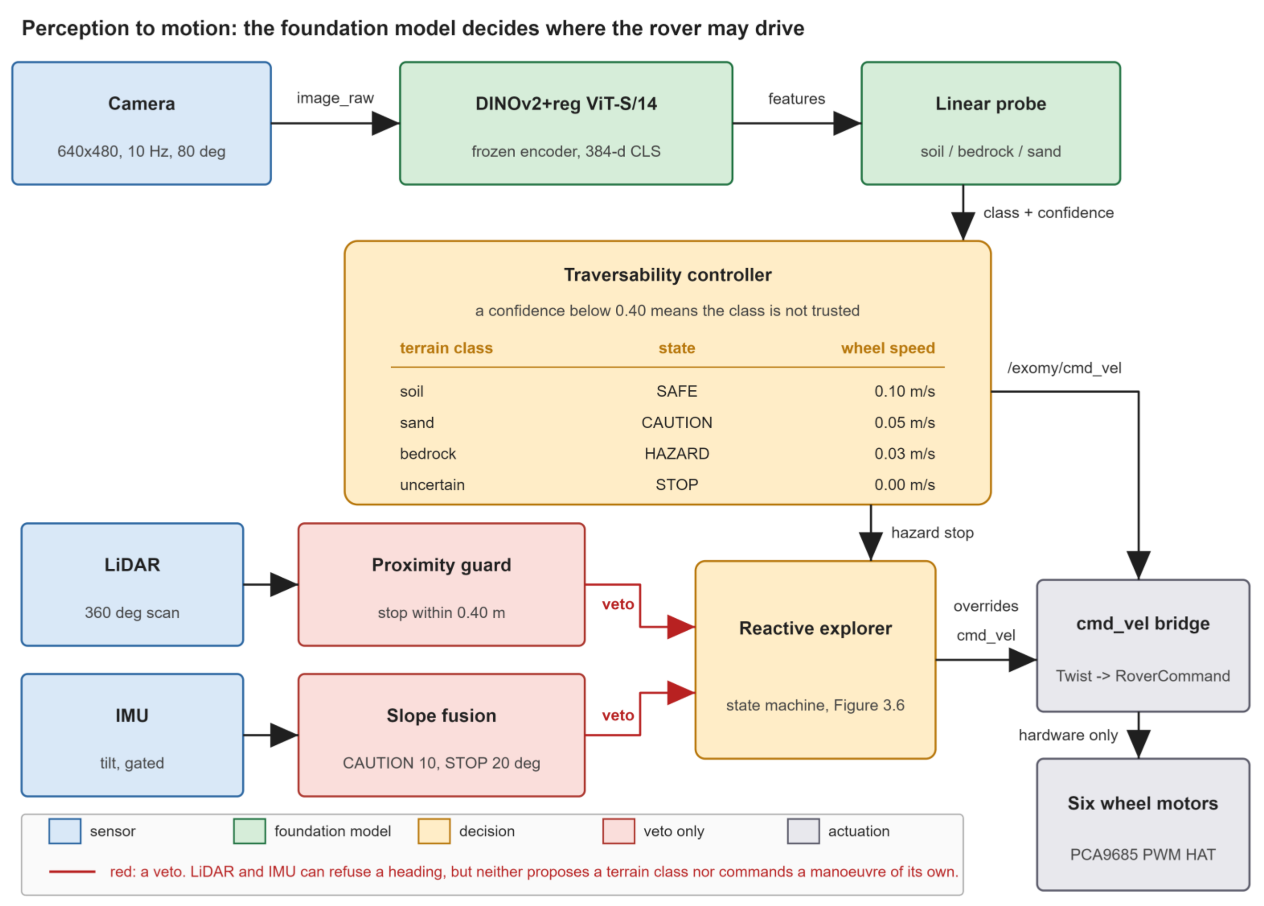
- Earth is **3 to 22 minutes** away by radio. A Mars rover must judge unfamiliar ground on its own.
- Current rovers rely on stereo geometry and hand-tuned traversability rules, which need mission-specific calibration and GPU training.
- This work asks whether a **frozen** pretrained vision model can replace that pipeline, and which pretraining objective transfers best to Mars terrain.

Aim and objectives

- Benchmark 22 variants across six pretraining paradigms on AI4Mars, every encoder frozen.
- Deploy the strongest encoder that fits a Raspberry Pi 4, inside a ROS2 stack.
- Turn class and confidence into a conservative speed policy, in Gazebo and on a real ExoMy rover.

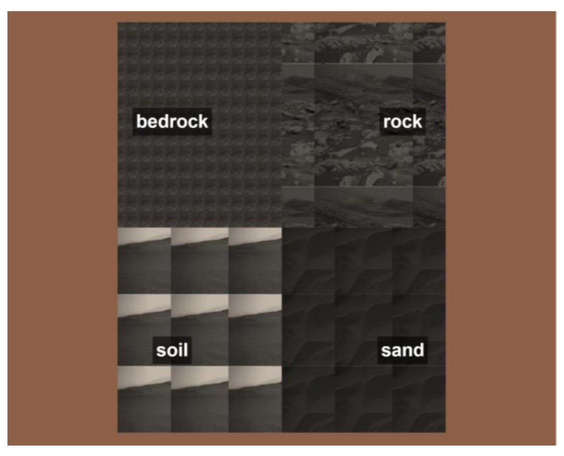
Method

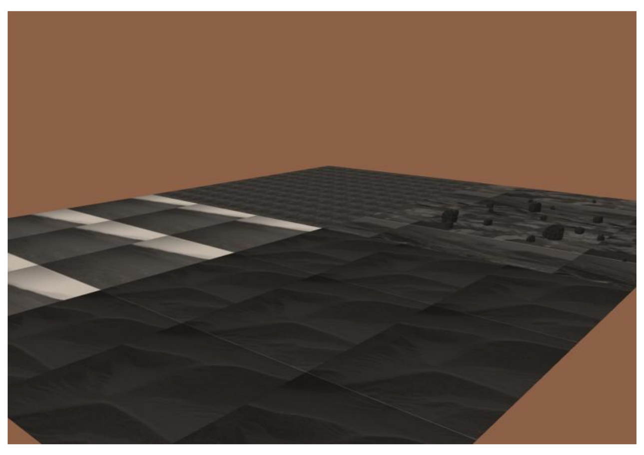
- Nothing is retrained. A frozen DINOv2 backbone produces the features; only a small linear probe is fitted on top.
- Class and confidence set the wheel speed. A confidence below 0.40 becomes a stop.
- LiDAR and IMU are vetoes only. Neither proposes a terrain class.



Camera to motion. The one trained component is the linear probe.

Simulation

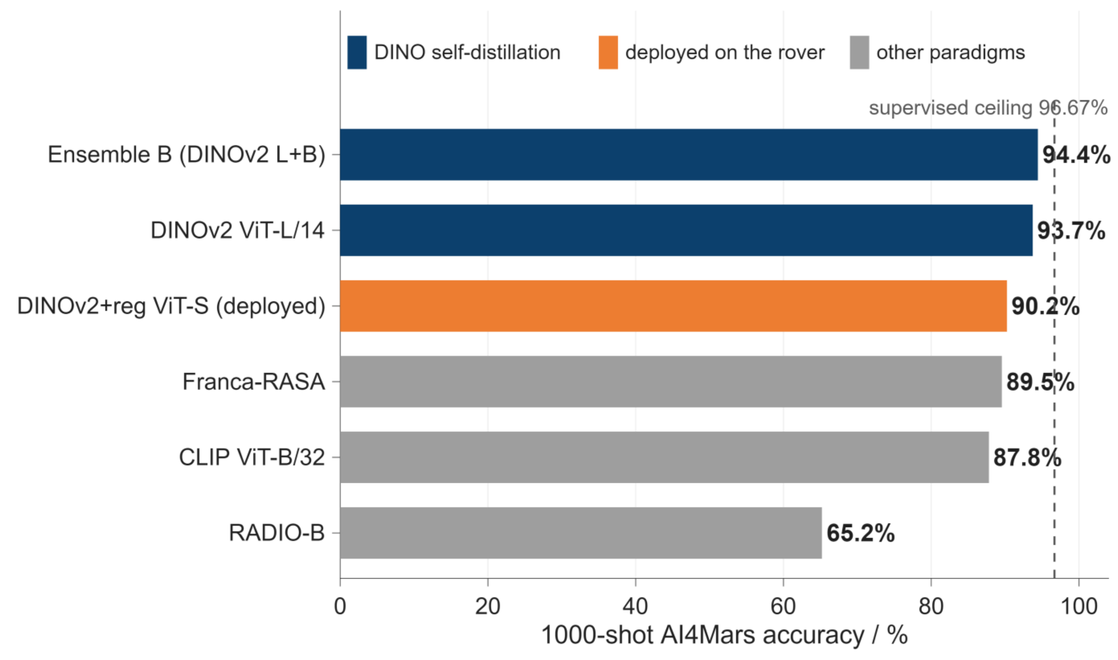




A 30 x 24 m Gazebo world with four AI4Mars-textured quadrants, used to test the speed policy on known ground before it ever reached the rover.

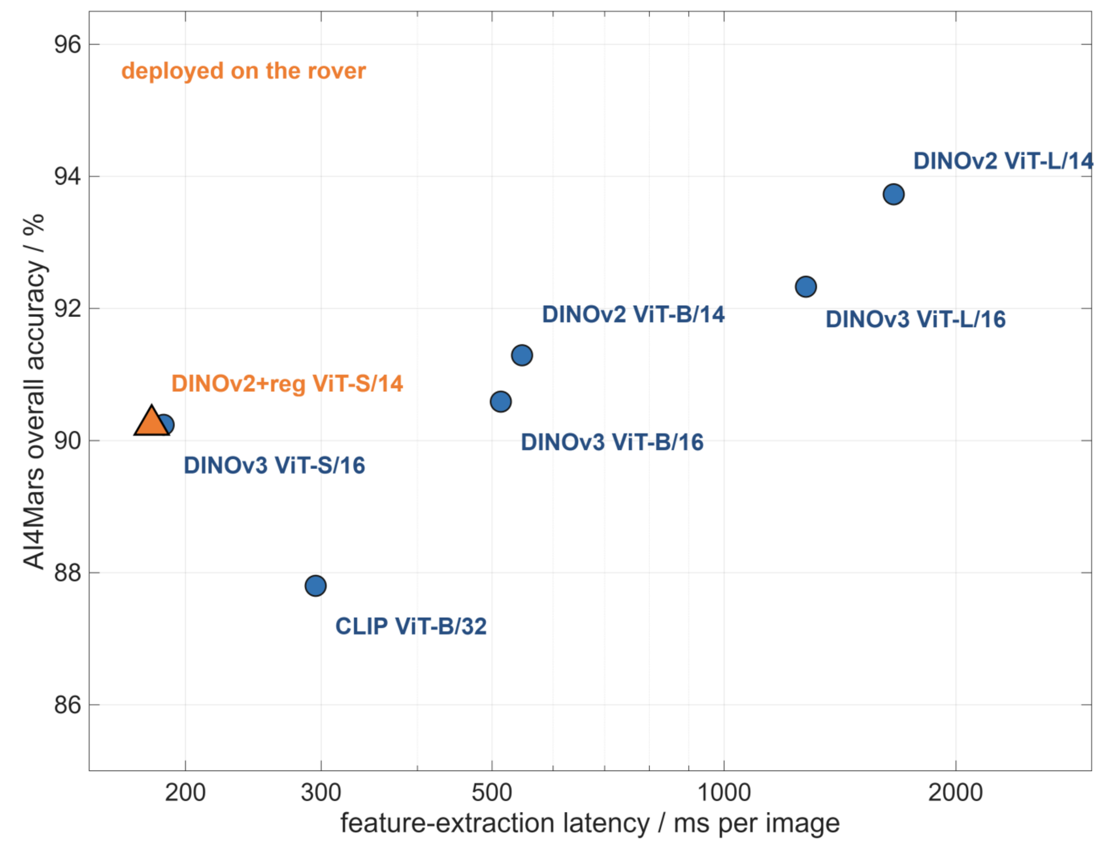
Model comparison

- DINO self-distillation wins. The top eleven of 22 variants are all DINO encoders or ensembles; every rival paradigm ranks below them.
- Curated data beats more data. DINOv2's 142M curated images beat DINOv3's 1.69B web-crawled images at ViT-B and ViT-L.



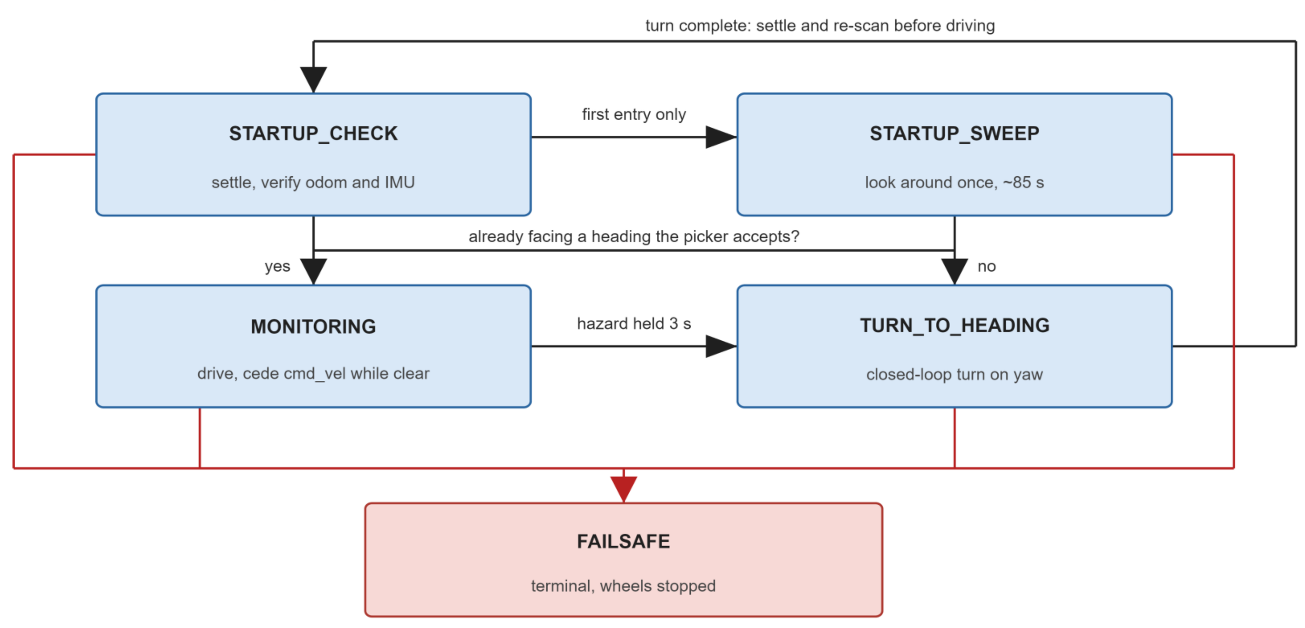
Six of the 22 variants: the two best overall, the model deployed on the rover, and the strongest entry from each of three rival paradigms.

Deployment trade-off



DINOv2 ViT-L scores 3.5 points higher, but it is 14 times larger and an order of magnitude slower. Giving up those 3.5 points is what buys a model that fits the rover's compute and memory budget at all.

Autonomy behaviour



The shipped state machine. There is no reverse gear: with no rear-facing sensing, every route that ends badly ends in a stop.

Results and conclusions

94.43%	**90.24%**	**570 ms**	**5 of 5**	**10 of 10**
best frozen-encoder accuracy	deployed on the rover	per image, Raspberry Pi 4	Gazebo safety compliance	real-hardware safety rounds

1. **Frozen features are enough to start.** A two-encoder DINOv2 ensemble comes within 2.24 points of a supervised specialist trained on the full labelled set, with no gradient update to the encoder at all: only a small linear probe is fitted.
2. **Conservative design absorbs classification error.** The deployed model named the terrain correctly at 3 of the 5 Gazebo positions, yet every one of the 5 produced a safe outcome: a confidence below 0.40 is converted into a stop, so an uncertain reading costs time rather than safety.
3. **Transfer is class-dependent, and the gap is a labelling gap.** On the rover's own camera, soil is classified perfectly and sand holds at 73.9%. Big rock is not in the deployed three-class label set, so all 36 photographs are answered as soil or sand at high confidence and the 0.40 stop threshold never fires. Trained balanced, the same frozen features separate big rock at 70.0%, so the remaining work is data and label-set design, not a stronger encoder.